

Hypo-Stage: Tool-Supported Architectural Hypothesis Engineering in Professional Software Development Teams — A Case Study

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Master's Thesis Defense

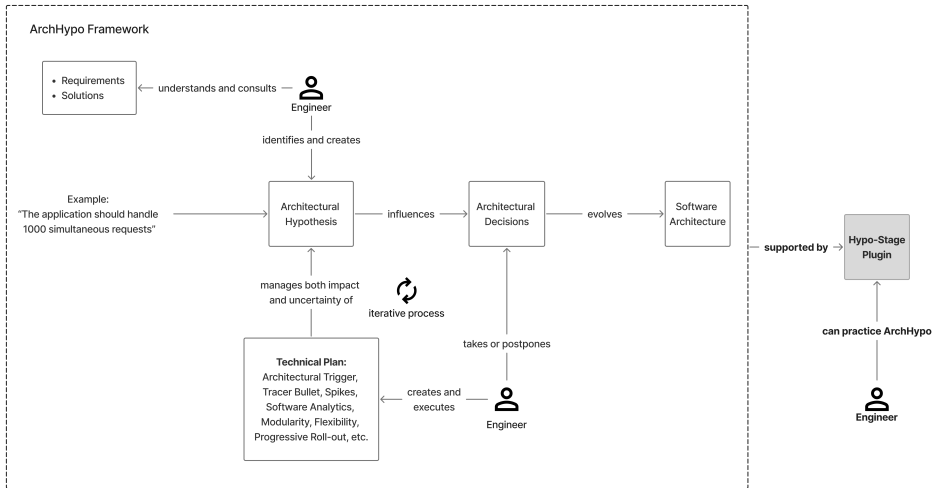
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Context



Problem & research questions

The Problem

- Deriving architectures still depends on architects' **tacit knowledge** and lacks **tool support** (Souza et al., 2019).
- ArchHypo structures uncertainty, but teams perceive it as **hard to learn** and missed guidance via supporting tools (Silva, Melegati, Silveira, et al., 2025; Silva, Melegati, Wang, et al., 2024).
- Hypo-Stage gives the initial tool support, but **impact on real teams** and refinements remained **unknown** (Corrêa, 2025).

Research questions

RQ1. How do teams *use* Hypo-Stage and ArchHypo on a scaled product?

RQ2. What *uncertainties* and recurring *patterns* do they record?

RQ3. What *benefits and challenges* do team members perceive?

RQ4. Which *refinements* to the tool emerge, and how do they address the challenges?

Method & study setup

Environment

Single organization

Scaled product

Backstage IDP

Teams

Product team

12 engineers

Platform team

4 engineers

Timeline

5 sprints
(≈ 9 weeks)

Sprint 1

Sprint 2

Sprint 3

Sprint 4

Sprint 5

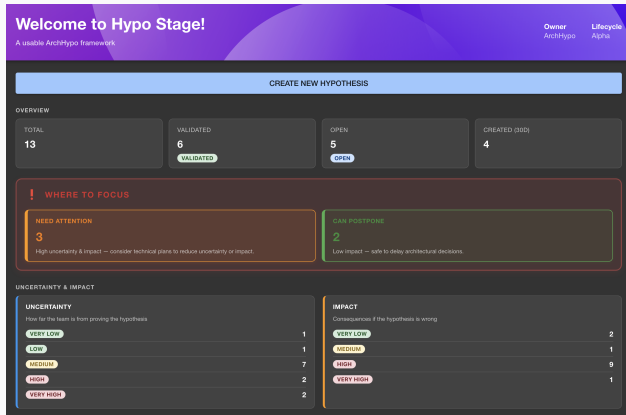
Artifact analysis
(13 hypotheses)

Data sources

triangulated

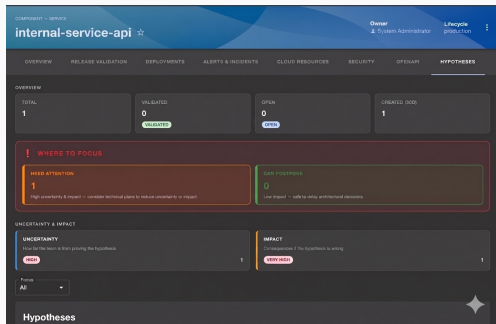
Participant
observation

Active-listening
sessions



Adoption was **senior-led**
and **event-anchored**:

- 12 of 13 hypotheses authored by senior/staff engineers
- Registration clustered around protected technical sessions
- Backstage integration cut context-switching cost

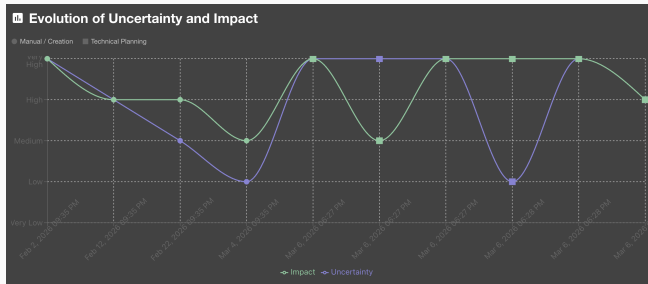


Uncertainty $\times$ impact, surfaced per service in the developer portal. Dominant quality attributes: **Reliability** and **Auditability**.

Five emerging hypothesis patterns

- **Hypothesis Cluster** — one platform concern $\rightarrow$ linked hypotheses
- **Regulatory Trigger** — compliance obligation as tracked uncertainty
- **Context-Agnostic Restructuring** — whether/how to decouple boundaries
- **Load-Validated** — proven under load, implementation still open
- **Shared Resource Governance** — ownership rules for shared infrastructure

First empirically grounded answer to *what shapes* architectural hypotheses take in practice.



Evolution chart: uncertainty & impact tracked across successive assessments.

The "Create New Hypothesis" form includes the following sections:

- Entity References:** A dropdown menu with a search bar and a "Please select at least one entity reference. Type to search existing components." prompt.
- Hypothesis Statement:** A text area with a "9/500 characters" limit.
- Source Type:** A dropdown menu.
- Uncertainty Level:** A section with a question "How far is the team from validating/falsifying this hypothesis?" and a scale from 1 to 5, with "Not rated" as an option.
- Impact Level:** A section with a question "What is the potential impact on the system if the hypothesis proves false?" and a scale from 1 to 5, with "Not rated" as an option.
- Quality Attributes:** A section with a question "Please select at least one quality attribute that this hypothesis affects. Click the field to open the list; each option shows a short concept in parentheses." and a dropdown menu.

Inline formulation guidance in the creation form lowers the falsifiability barrier.

9 refinement categories — including prioritization, the evolution chart, and form usability — each traced to an observed friction point.

Implications & lessons learned

Finding

Even with tool support, hypothesis work stayed **senior-anchored** (12 of 13).

Implication

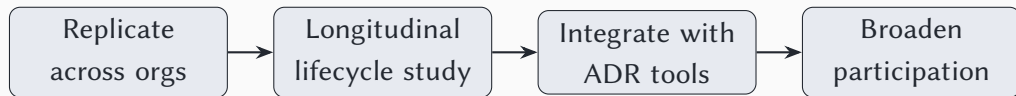
A tool removes **friction** but cannot supply the **organizational conditions** for broad participation.

Lesson

Ship hypothesis support **inside** the developer portal — *extend* the workflow, don't add a new one.

Distributed architectural work needs lightweight structure that preserves team autonomy while protecting shared decision quality.

Future work & closing



Takeaway

Tool-mediated ArchHypo succeeds at **integration and traceability**; the remaining challenge is **organizational and pedagogical**.

Limitations: single organization, 13 hypotheses, ≈ 9 weeks, researcher as participant — emerging propositions, not consolidated findings.